KINGDOM OF SAUDI ARABIA

PHARMA-BRIDGE

An Integrated Applied R&D and Smart Manufacturing Platform

> Bridging the structural gap between clinical/academic research and large-scale pharmaceutical manufacturing.

GENERIC PHARMA | PRIVATE LABEL | VISION 2030

STRATEGIC PARTNERS

NAMA | Majour | Nana | Waraqa | Integra

PHARMA-BRIDGE

A first-of-its-kind industrial activation platform in Saudi Arabia.

- Bridging Clinical Research & Manufacturing
- Accelerating TRL 7-8 Transition
- Empowering Physician Innovators

Vision 2030

DIRECT STRATEGIC ALIGNMENT

Supporting localization of pharmaceutical R&D and drug security.

Executive Summary

VISION 2030

Transforming Scientific IPs into Market-Ready Products

The Core Mission

To bridge the structural gap between academic research and large-scale manufacturing. PHARMA-BRIDGE transforms scientific concepts into registrable products, solving the "valley of death" in drug development.

Strategic Ecosystem

Built on a foundation of operational excellence by leveraging existing assets:

- NAMA (Clinical)
- Majour (Procurement)
- Nana (Logistics)
- Integra (AI)

Key Deliverables

- ✓ Small-batch validation & pilot runs
- ✓ SFDA-compliant BE & Phase I studies
- ✓ "Ready-to-Scale" technical dossiers

National Impact

Increases "True Local Content" by owning the formulation IP, not just the packaging. Significantly reduces innovation risk for manufacturers and shortens time-to-market for new generic and private label products.

Strategic Foundation

Leveraging operational excellence and established partnerships

REGULATORY READINESS

NAMA Medical

Utilizing SFDA-licensed Clinical Trials Monitoring Center (License: CT-2025-DR-0002) for regulatory compliance and study monitoring.

ACADEMIC & CLINICAL BACKING

KSU & KFSHI

Strategic MOUs with King Saud University and King Faisal Specialist Hospital for bioequivalence studies and R&D support.

LOGISTICAL INFRASTRUCTURE

Nana Healthcare

Integration with multi-tenant healthcare platform for rapid market access and specialized cold-chain distribution.

EVIDENCE & PUBLICATION

Waraqa

Dedicated support for scientific data publishing and generation of local clinical evidence to support market adoption.

AI & DATA MODELING

Integra

Using Integra's capabilities to optimize RCM (Regulatory Change Management) and data-driven formulation modeling.

Platform Architecture: The Five Pillars

A synchronized ecosystem connecting innovation to industrial scale

PHARMA BRIDGE

Industrial Catalyst & Innovation Core

1 Applied R&D & Formulation Hub

Transforming molecules into stable, optimized formulations and "Private Label" products.

Output: Optim

2 Smart Pilot Manufacturing

Small-batch production for Clinical & Market Test Batches. Bridging to TRL 7-8.

Output: Clinical Batches

3 Clinical Validation (NAMA Layer)

Regulatory-grade BE studies and Phase I trials ensuring SFDA/ICH-GCP compliance.

Output: SFDA Compliance

4 Market Integration (Majour Layer)

Aligning procurement with patient outcomes via value-based reimbursement models.

Output: Reimbursement

5 Distribution & Access (Nana Layer)

Overcoming "Last-Mile" barriers through digital demand aggregation and logistics.

Output: Market Access

01

Applied R&D & Formulation Hub

Transforming raw molecules into stable, optimized formulations ready for the market.

TRL 4-6 Activation

Core Objective

To develop "Private Label" products for specialized clinics and high-demand generics based on real-time market gap analysis, bridging scientific theory with industrial practicality.

Key Functions

- **Formulation Optimization:** Developing stable chemical compositions tailored for local climate conditions and supply chains.
- **Market Gap Analysis:** Real-time identification of shortage drugs and high-value therapeutic classes.
- **Private Label Development:** Custom formulations for dermatology and aesthetic clinics (Physician-Led Innovation).

INTEGRA

AI-Powered Modeling: Utilizing Integra's technological capabilities to optimize RCM (Regulatory Change Management) and predict formulation stability, reducing trial-and-error cycles.

Robust Formulations

CMC Documentation

Pilot Specs

02

Smart Pilot Manufacturing Unit

Small-batch, high-flexibility production bridging the critical gap between lab scale and commercial manufacturing.

TRL 7-8 Activation

Core Objective

To enable physicians and researchers to produce "Clinical Batches" and "Market Test Batches" without the immense overhead of mega-factories, effectively removing the barrier to entry for local pharmaceutical innovation.

Capabilities

- **Agile Production Lines:** GMP-aligned pilot lines designed for rapid changeovers between different formulations.
- **Scale-Up Readiness:** Defining critical process parameters (CPP) to ensure smooth transfer to mass production.
- **Flexible Batch Sizes:** Supporting variable batch sizes suitable for Phase I/II trials and initial market seeding.

THE INDUSTRIAL BRIDGE

Serves as the vital link between academic concepts (TRL 3-4) and industrial commercialization (TRL 9). By validating manufacturability early, we significantly de-risk the investment for large-scale manufacturers.

Clinical Batches	Market Test Units	Tech Transfer Pkg

03

Clinical & Bioequivalence Validation

The NAMA Layer: Ensuring regulatory-grade human studies for full SFDA compliance.

SFDA & ICH-GCP

Core Objective

To execute regulatory-grade human studies that validate safety and efficacy, acting as the critical bridge between formulation and market authorization.

Assets & Capabilities

- **40-Bed Clinical Unit:** A specialized facility designed for high-throughput Bioequivalence (BE) studies and Phase I trials.
- **Regulatory Alignment:** Full adherence to SFDA and ICH-GCP standards, ensuring data acceptance.
- **Accelerated Timeline:** Significant reduction in validation phase duration by localizing studies instead of outsourcing abroad.

NAMA MEDICAL

Licensed Monitoring Center: Leveraging NAMA's SFDA license (CT-2025-DR-0002) to generate high-quality local clinical evidence that supports national drug security.

BE Reports

Regulatory Dossiers

Safety Data

04

Value-Based Market Integration

Ensuring products are not just manufactured, but effectively integrated into the healthcare spending value chain.

Core Objective

To align pharmaceutical procurement with patient outcomes and cost efficiency, moving beyond simple volume-based supply to value-based care models.

Key Functions

- **Outcome-Based Models:** Linking product usage to patient health metrics for smarter reimbursement.
- **Procurement Alignment:** Connecting manufacturing output directly to hospital demand signals and budget cycles.
- **Real-World Evidence:** Generating post-market data to support pricing and formulary inclusion.

MAJOUR

SaaS Enablement: Utilizing Majour's platform to track product performance and automate outcome-based reimbursement, ensuring sustainable market adoption.

Value Dossiers

Contracting Frameworks

Outcomes Dashboards

05

National Distribution & Access

Overcoming the "Last-Mile" and warehousing barrier via specialized digital logistics.

The Nana Layer

Core Objective

To ensure immediate commercialization and patient access by leveraging digital demand aggregation and a specialized medical logistics infrastructure, solving the critical "last-mile" challenge.

Key Functions & Value

- **Last-Mile Logistics:** Robust cold-chain distribution network reaching pharmacies and clinics nationwide.
- **Digital Demand Aggregation:** Real-time visibility into market needs to drive efficient inventory placement.
- **Rapid Commercialization:** Seamless transition from manufacturing release to market availability, minimizing shelf-time.

Nana

Digital Logistics Platform: Integration with Nana's multi-tenant healthcare platform enables instant channel activation and transparent inventory tracking from factory to patient.

Channel Activation

Service-Level KPIs

Fulfillment Analytics

Operating Model: An "Industrial Catalyst"

Connecting Innovation to Industrial Scale

INPUTS

Scientific IP

Clinical concepts from physicians & academic research (NAMA/KSU).

Market Gaps

Real-time shortage data & demand analysis (Majour/Nana).

PHARMA-BRIDGE Asset-Light Core

Innovation Arm for Saudi Pharma

Applied R&D
Formulation & Optimization

Pilot Manufacturing
Small Batch Production

Validation
Bioequivalence & Clinicals

VALUE OUTPUTS

FOR PHYSICIANS

Turn-Key Innovation

We act as their R&D department to launch private-label dermatology or therapeutic lines.

FOR MANUFACTURERS

De-Risked Scale-Up

We provide "Ready-to-Scale" dossiers and validated tech transfers, reducing innovation risk.

FOR THE REGULATOR (SFDA)

National Drug Security

We provide high-quality local data and localized R&D that supports national sovereignty.

Technical Impact & National Alignment

Strategic Contributions to Saudi Arabia's Pharmaceutical Security

VISION 2030 ALIGNED

R&D Localization

Directly contributes to Vision 2030 by moving beyond "packaging localization" to owning the IP and formulation development process within the Kingdom.

National Drug Security

Reducing reliance on imported finished products by establishing local capabilities for critical generics and private label manufacturing.

Knowledge Economy

Transforming academic papers from KSU/KFSHI into tangible industrial assets through the Waraqa-supported evidence cycle.

Indicative KPI Targets

Projected impact metrics for the first 24 months of operation.

50%

FASTER TIME-TO-MARKET

Reduction in development cycle vs. traditional outsourcing models.

30%

LOCAL CONTENT UPLIFT

Increase in "True Local Content" score for partner manufacturers.

5+

ANNUAL BE STUDIES

Executing local bioequivalence studies via NAMA Bio-Center.

8

DOSSIERS / YEAR

Ready-to-scale technical dossiers transferred to local factories.

Pilot Phase Roadmap

24-Month Execution Plan: From Setup to Commercial Scale

Months 1 - 24

MONTH 1 - 6

Foundation Setup & AI Integration

- **Smart Pilot Lab:** Installation of equipment and validation of SOPs.
- **AI Tooling:** Integration of "Integra" market-gap analysis tool.
- **Quality Systems:** Establishing QC baselines aligned with SFDA.

MONTH 7 - 12

Development & Innovator Onboarding

- **Physician Cohort:** Onboarding first group of Physician-Innovators.
- **Formulation R&D:** Development of private-label skin/therapeutic lines.
- **Prototyping:** Production of initial market test batches.

MONTH 13 - 18

Clinical Validation & Bioequivalence

- **BE Studies:** Execution of 3 Pilot Bioequivalence studies via NAMA.
- **Data Packages:** Compilation of regulatory dossiers for submission.
- **Pilot Review:** Assessment of TRL 7-8 readiness.

MONTH 19 - 24

Tech Transfer & Commercial Launch

- **Scale-Up:** Tech-transfer to local large-scale manufacturers.
- **Market Entry:** Commercial launch via Nana distribution network.
- **Value Contracts:** Initiation of outcome-based agreements with Majour.

Infrastructure R&D Operations Regulatory Commercialization

PHARMA-BRIDGE
Realizing the Vision

> More than a facility, PHARMA-BRIDGE is a scalable industrial engine. We synchronize the clinical expertise of NAMA, the logistical power of Nana, and the value-based vision of Majour to ensure Saudi innovation becomes a global pharmaceutical reality.

NAMA Medical | Nana Logistics | Majour

Partner With Us

Join us to prioritize molecules, co-develop private labels, and localize high-impact generics.

1. **Alignment Workshop**
   Strategic session within 2 weeks to identify shared goals.
2. **Candidate Selection**
   Feasibility analysis for molecule or private label line.
3. **Regulatory Scoping**
   Defining the pathway and timeline for SFDA approval.

partnerships@pharma-bridge.sa

Reach out today to initiate the collaboration.